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Paleomagnetic Evidence for the Timing of Collapse and Resurgence of the Lake City Caldera, San Juan Mountains, Colorado

RICHARD L. REYNOLDS, MARK R. HUDSON,[1] AND KEN HON

U.S. Geological Survey, Denver, Colorado

Rocks of the 23.1-m.y.-old Lake City caldera consist of the compositionally zoned Sunshine Peak Tuff, postcollapse intracaldera lava flows, and resurgent quartz syenite intrusions. Declinations of reversely magnetized ($I = -45°$ to $-75°$) Sunshine Peak Tuff change from easterly ($D = 93°$–$130°$) throughout most of the tuff to southerly ($D = 195°$–$207°$) within the late eruptive phases. The postcollapse lava flows are also reversely magnetized, and their emplacement appears to have spanned resurgence of the caldera. In contrast, the uppermost levels of the resurgent intrusion record two distinct components of magnetization: a reversed component isolated by thermal demagnetization typically at temperatures greater than 400°C and a normal component defined by thermal demagnetization at lower temperatures. A baked-contact test implies that emplacement and initial cooling of the intrusion occurred during a time of reversed polarity and that continued cooling at lower blocking temperatures took place during normal polarity. Assuming that the reversed directions of the extrusive and intrusive rocks record a single period of reversed polarity as suggested by the paleomagnetic and radiometric age data, the time for caldera development (from eruption of the ash flow tuffs to emplacement of the resurgent intrusion) was less than about 300,000 years on the basis of best estimates of durations of reversed polarities in the early Miocene. The estimated 300,000-year time span for the development of the Lake City caldera is very close in duration to the radiometrically determined time spans of much younger (Pleistocene) calderas having similar dimensions and similar volumes of erupted material.

INTRODUCTION

Few large resurgent calderas contain exposures of sufficiently complete stratigraphic and intrusive sequences to allow accurate determination of age relations among all events of the caldera cycle. The timing of eruptive events associated with young calderas are known from conventional potassium-argon (K-Ar) ages of the ash flow tuffs and associated postcollapse volcanic rocks, whose relation to the timing of resurgence can be determined from geologic observations [*Bailey et al.*, 1976; *Hildreth et al.*, 1984; *Christiansen and Blank*, 1972; *Christiansen*, 1979, 1982, 1984; *Doell et al.*, 1968]. Thus the lifespan of calderas can be estimated, within the precision of the K-Ar analyses, from the eruption of ash flow tuffs and caldera collapse to resurgence, accompanied by the emplacement of large intracaldera intrusions. Even in young calderas, however, the precision of K-Ar analyses cannot discriminate rapid events, such as the eruption of multiple units within an ash flow sheet [*Mahood and Drake*, 1982; *Mahood*, 1981; *Hildreth et al.*, 1984; *Hildreth and Mahood*, 1985].

The Lake City caldera in southwestern Colorado (Figure 1) provides an excellent opportunity to examine timing relations within a well-exposed resurgent caldera [*Smith and Bailey*, 1968; *Lipman et al.*, 1973; *Lipman*, 1976*a*, *b*, 1984] where deep erosion has exposed the top of a large resurgent intrusive body. A remarkable stratigraphic sequence, consisting of compositionally zoned intracaldera ash flow tuff (the Sunshine Peak Tuff) capped by postcollapse lavas, also is preserved within the caldera [*Hon et al.*, 1983*a*; *Lipman*, 1976*a*] (Figure 2). Unfortunately, the age (23 m.y.) of the Lake City caldera precludes the use of K-Ar ages to discriminate individual events within the caldera cycle [see *Hildreth and Mahood*, 1985]. This is a problem common to other well-exposed calderas [*Lipman*, 1983; *Lambert*, 1974; *Fridrich and Mahood*, 1984; *Hildebrand*, 1984] because the degree of exposure is often a direct function of age and tectonic setting. Nevertheless, results of radiometric dating constrain the evolution of the Lake City caldera to a relatively short interval of time.

Potassium-argon determinations on six biotite and five sanidine separates from rocks of the Lake City caldera yield an average age of 23.1 m.y. with an error of ± 1.0 m.y. (2σ). The mean ages of the Sunshine Peak Tuff, the postcollapse lava flows, and the intrusions are all concordant and range from 22.2 to 24.0 m.y. Three of the samples were suspected of having retained argon; the remaining eight samples give a tighter range of mean ages from 22.7 to 23.6 m.y. and yield an average age of 23.2 ± 0.8 m.y. (2σ). (We have included all analyses in determining the average age of 23.1 ± 1.0 m.y. for the igneous activity related to the Lake City caldera.) Although the mean ages from individual mineral separates do not reflect the relative timing of igneous events, the similar K-Ar ages and geologic data do document that the ash flow tuffs, the postcollapse lavas, and the resurgent intrusions represent distinct phases of a single caldera [*Hon et al.*, 1983*a*]. A direct relation between extrusive and intrusive activity in large collapse calderas has been convincingly demonstrated in only a few other studies [*Fridrich and Mahood*, 1984; *Lipman*, 1983, 1984; *Lambert*, 1974].

Paleomagnetic methods can assist field, petrologic and geochemical studies in deciphering the histories of calderas [*Dalrymple et al.*, 1965; *Doell et al.*, 1968; *Reynolds*, 1977; *Hagstrum et al.*, 1982]. With the exception of the work by *Hagstrum et al.* [1982] on the late Oligocene Questa caldera, these detailed paleomagnetic studies have investigated young (Pliocene and Pleistocene) calderas. In this paper we report the results of a paleomagnetic study undertaken to constrain the relative timing of the eruptive and intrusive events associated with the Lake City caldera with greater resolution than

[1] Also at Department of Geology, Colorado School of Mines, Golden.

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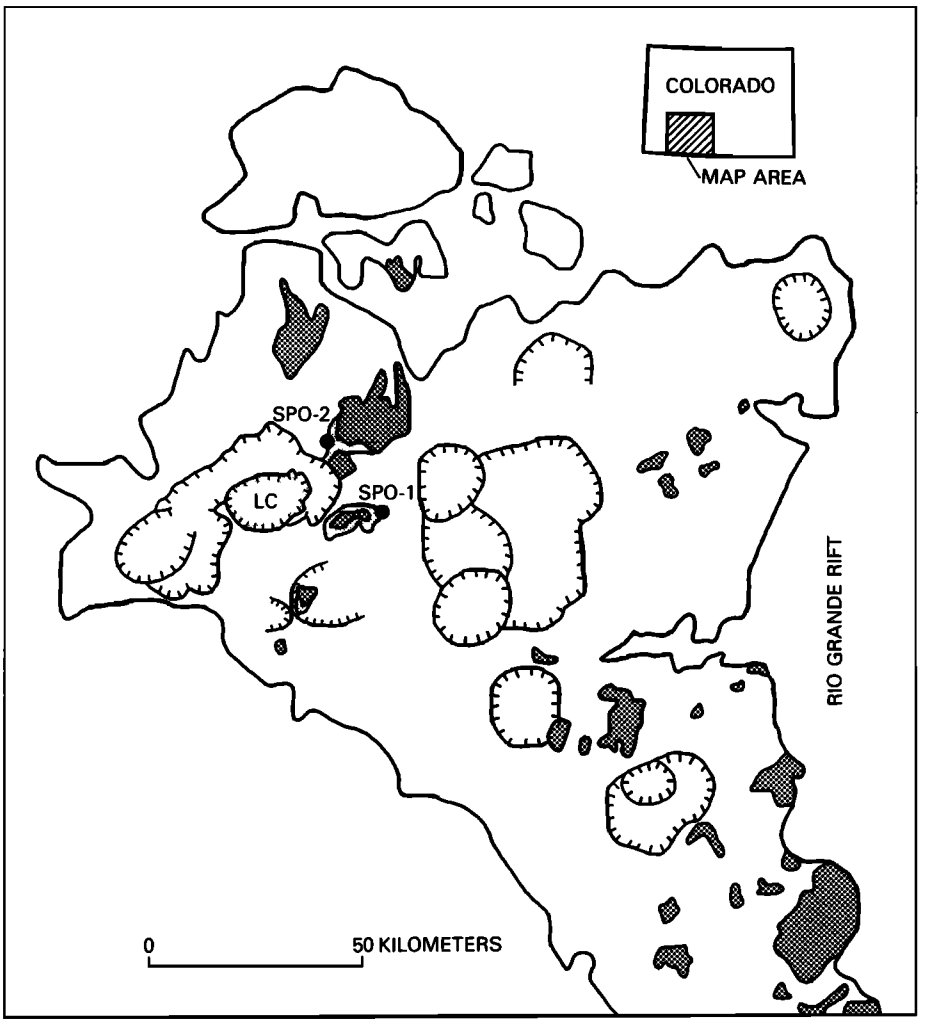


Fig. 1. Map of the San Juan volcanic field, southwestern Colorado (modified from *Steven and Lipman* [1976]) showing Oligocene calderas (hachured lines), the Miocene Lake City caldera (LC; lightly stippled area within hachured line); outflow facies of the Sunshine Peak tuff (lightly stippled areas beyond the Lake City caldera); sampling sites (SPO-1, 2) in the outflow facies; and areas of Miocene basalt and basaltic andesite (shaded).

possible by K-Ar dating methods. Although paleomagnetic data do not provide absolute ages, as do radiometric techniques, in this case it was possible to estimate relative time based on models of magnetic secular variation and on reasonably well-established records of polarity reversals of the earth's field. The individual units of the Sunshine Peak Tuff were studied primarily to determine whether their eruption and cooling occurred over long or short periods of time, relative to directional changes of the earth's paleomagnetic field. Paleomagnetic data were also obtained to aid in determining the time of resurgence relative to emplacement of lava flows and to assess whether exposures of the resurgent intrusions are part of one large intrusive body, as implied by geological, geochemical, and geophysical data, or alternatively represent separate bodies having different intrusive and cooling histories.

Finally, we considered the idea that the magnetic polarities of the units related to the Lake City caldera might be used to estimate crudely the lifespan of the caldera. For example, a single polarity chron recorded during volcanism and initial cooling of the resurgent intrusion (a possibility suggested herein for the Lake City caldera) would imply a short lifespan relative to the frequency of polarity changes in the early Miocene. Although it is not possible, using K-Ar ages, to correlate the polarities of rocks more than about 5 m.y. old to individual polarity chrons [*Irving and Pullaiah*, 1976], it may be useful, in conjunction with radiometric ages, to compare such polarities (whose relative ages are known) to parts of the Cenozoic polarity time scale based on marine magnetic anomalies. In some cases, this comparison may constrain the time interval represented by the polarity data. The utility of this approach is limited by a number of factors: (1) variable durations of polarity chrons, (2) the chance that episodic igneous activity within a caldera cycle occurred during more than one chron of the same polarity while "skipping" intervening times of the opposite polarity, and (3) the possibility that the igneous activity encompassing the development of the caldera is not sufficiently represented, either because of inadequate sampling or because of incomplete records caused by erosion or lack of exposure. The results from the Lake City caldera illustrate some of the uses and limitations of paleomagnetism to improve understanding of the development of calderas.

GEOLOGIC SETTING

The 23.1 m.y. Lake City caldera is the youngest caldera within the middle Tertiary San Juan volcanic field of southwestern Colorado [*Steven and Lipman*, 1976; *Mehnert et al.*, 1973] (Figure 1). Unlike the predominantly intermediate-

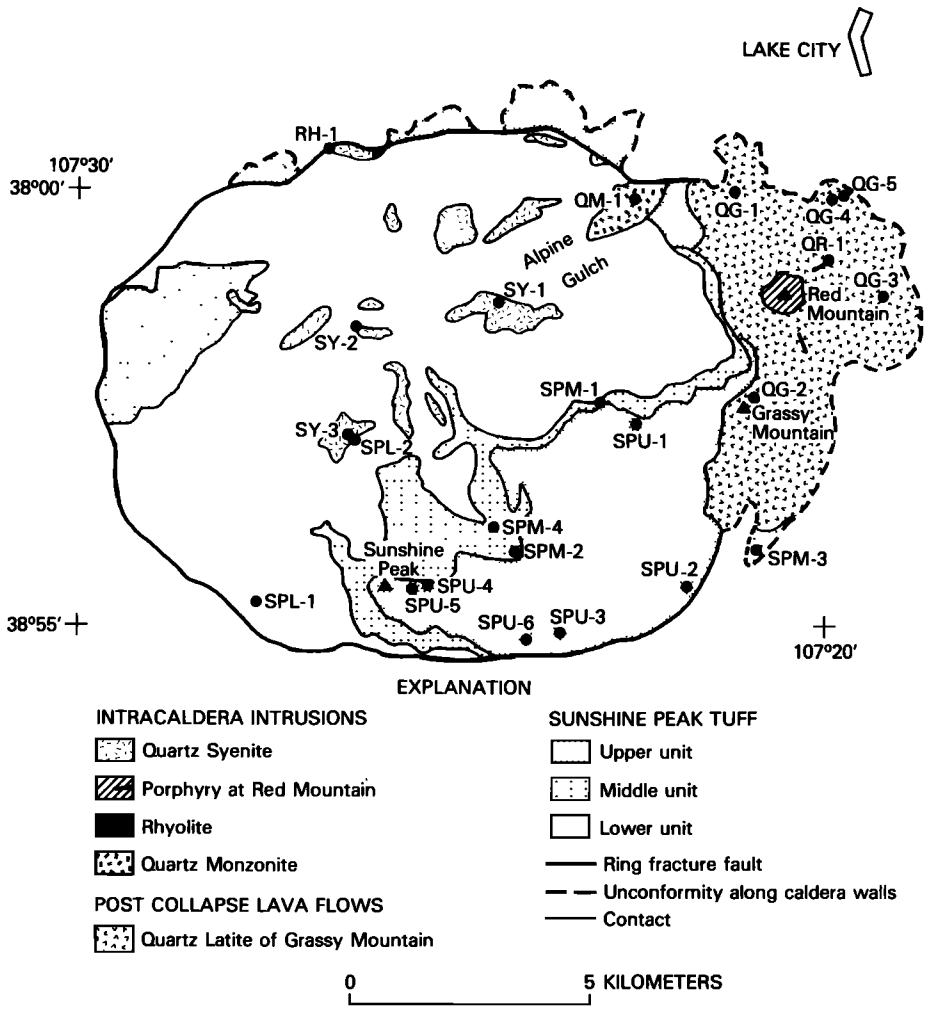


Fig. 2. Generalized geologic map of the Lake City caldera (modified from K. Hon, unpublished data, 1984), San Juan Mountains, southwestern Colorado, showing locations of sampling sites.

composition rocks characteristic of the earlier Oligocene volcanism in the region, rocks of the early Miocene Lake City caldera are closely related to a bimodal assemblage of alkali basalt and rhyolite that is generally associated with post-Oligocene igneous activity in southwestern Colorado [*Lipman et al.*, 1978] as well as throughout much of the western United States [*Christiansen and Lipman*, 1972].

The Lake City caldera formed in response to eruption of the Sunshine Peak Tuff [*Lipman et al.*, 1973]. Much of the estimated 300–400 km^3 of ash flow material produced during this eruption became ponded within the caldera contemporaneously with caldera-collapse breccias that accumulated during subsidence of the caldera. Three compositionally distinct members of Sunshine Peak Tuff have been recognized within the caldera [*Hon et al.*, 1983*a*] (Figures 2 and 3). The earliest unit (lower Sunshine Peak Tuff) is a high-silica rhyolite (76% SiO_2) and is the only member found in the outflow facies beyond the rim of the caldera (Figure 1). The middle unit is also rhyolite (74% SiO_2), whereas the upper Sunshine Peak Tuff is a quartz trachyte (68% SiO_2) and is mainly preserved along the eastern and southern margins of the caldera (Figure 2). Mappable cooling breaks and partings occur in both the middle and upper sunshine Peak Tuff (Figure 3) and allow the subdivision of both the middle and upper members into lower and upper subunits (K. Hon, unpublished data, 1985). In general, no major changes in bulk rock geochemistry occur between the subdivisions of an individual member (K. Hon, unpublished data, 1985); however, the break within the upper member is an important stratigraphic marker for interpretation of paleomagnetic data.

Following caldera collapse, the upper Sunshine Peak Tuff and parts of the middle rhyolitic member were eroded along the eastern caldera margin before eruption of quartz latitic lava flows. Field relations indicate that emplacement of these postcollapse lavas bracket the time of resurgent doming of the Lake City caldera (K. Hon, unpublished data, 1985). Near Grassy Mountain on the southeast margin of the caldera (Figure 2), lava flows are separated by thin lenses of bedded volcaniclastic sediments that dip about 40°, parallel to the lavas. This relation indicates that lava flows in this area were originally horizontal and that tilting occurred during formation of the resurgent dome. Similar lava flows in the northeastern part of the caldera, however, lie on tilted Sunshine Peak Tuff along a subhorizontal erosional unconformity and clearly postdate resurgence.

The postcollapse lavas were intruded by a younger quartz latite porphyry at Red Mountain [*Bove*, 1984] (Figure 2). Red Mountain is the eroded core of a volcanic dome complex that is now largely alunitized as a result of syngenetic solfataric alteration [*Bove*, 1984; *Mehnert et al.*, 1979].

Within the central and northern portions of the caldera are isolated quartz syenite exposures that are thought to represent

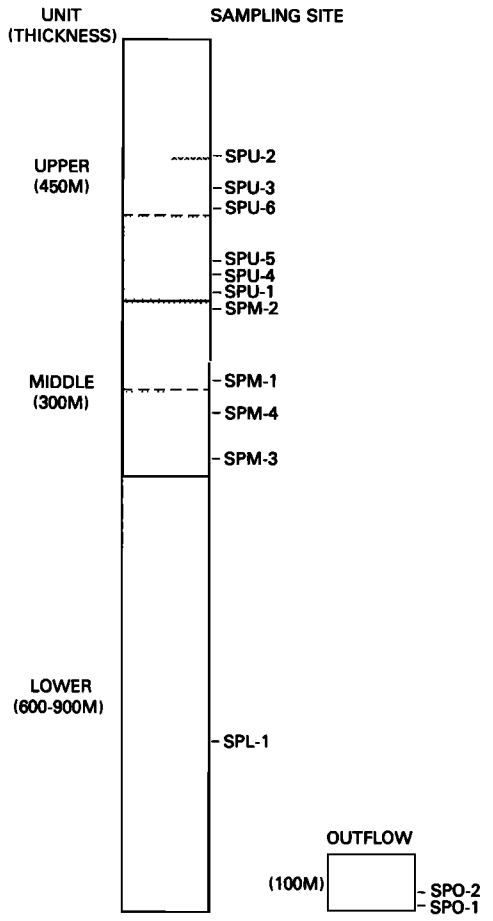


Fig. 3. Schematic stratigraphic section of the Sunshine Peak Tuff showing approximate positions of paleomagnetic sampling sites within intracaldera tuff (long column) and in outflow of the lower unit. Dashed lines denote partial cooling breaks.

the upper levels of the resurgent intrusion (Figure 2). Quartz syenite from these exposures is petrographically and geochemically similar, suggesting that they are all part of a single body. Two other intrusions were sampled: a 22.8 ± 0.9 m.y. quartz monzonite pluton (60–68% SiO_2) in the northeast part of the caldera (site QM-1, Figure 2), and an undated high-silica rhyolite ring fracture dike found along the northern caldera margin (site RH-1, Figure 2). Contrasts in texture, trace element content, and isotopic composition indicate that the quartz monzonite is not part of the main resurgent intrusion (K. Hon and Z. Peng, unpublished data, 1985). The quartz monzonite is thought to postdate the quartz syenite because the quartz monzonite intrusion is not offset by a major fault along Alpine Glulch that appears related to emplacement of the resurgent quartz syenite. The rhyolite dike was emplaced into caldera fill megabreccia of the Sunshine Peak Tuff along the northern ring fault and was then apparently intruded by a body of quartz syenite.

Emplacement of the intrusions into the lower caldera fill sequence recrystallized the adjacent lower Sunshine Peak Tuff. The resurgent intrusion also apparently supplied heat for widespread propylitic alteration of the intracaldera Sunshine Peak Tuff. Propylitization is most intense in the lower Sunshine Peak Tuff and the least within the upper member.

Field and Laboratory Methods

Samples were collected either as block samples or as cores using a portable drill and were oriented by means of magnetic or sun compass. In the laboratory, one or two cylindrical specimens (~2.5 cm in diameter and length) were cut from the field cores or drilled from block samples. In this way, 196 specimens were obtained from 14 sites (119 samples) in the Sunshine Peak Tuff, 91 specimens from lava flows and quartz latite porphyry (6 sites, 52 samples), and 90 from intracaldera intrusions (5 sites, 66 samples).

Directions and intensities of remanent magnetization were measured using a 5-Hz spinner magnetometer. Most specimens were subjected either to alternating field (af) demagnetization in a tumbling degausser or to thermal demagnetization in a noninductively wound furnace. The af treatment of at least one pilot specimen from each site of ash flow tuff and lava flow involved demagnetization in 10–12 peak field steps from 3 to 90 or 100 milliteslas (mT); components of magnetization were resolved from orthogonal vector diagrams using vector subtraction. Because pilot samples exhibited univectorial decay in response to af cleaning above the 10 mT cleaning level, samples from these sites were routinely demagnetized at intervals of 10 mT typically from 10 to 60 or 70 mT, and the mean direction at the demagnetizing step which resulted in the least dispersion was taken to represent a site (Table 1). Thermal demagnetization (9–11 steps from 100° or 300°C to 680°C) of 60 specimens of tuff and lava flow was carried out to determine magnetic components carried by magnetite and hematite. Intrusive rocks were demagnetized at the same AF levels (typically, 12 steps from 3 to 80 mT; 44 specimens) and temperatures (11 steps form 100° to 680°C; 46 specimens) as used for extrusive rocks. Because the intrusions have more than one magnetic component, all af and thermal results are given as remanent magnetizations determined using vector subtraction.

Although we avoided sampling areas or exposures that had been obviously affected by hydrothermal alteration, we paid close attention to the possibility of later remagnetization of intracaldera rocks, particularly of the Sunshine Peak Tuff. Accordingly, we examined the mineralogy of selected samples as well as certain rock magnetic properties to help identify magnetic carriers and to complement the paleomagnetic results. The studies included reflected-light microscopic examination of at least one polished thin section from each site (a total of 30 sections), thermomagnetic analysis in a field of 0.18 T, and measurement of laboratory-induced isothermal remanent magnetization (IRM) in fields up to 0.7 T.

Results

Sunshine Peak Tuff

The lower member of the Sunshine Peak Tuff was sampled at two sites (SPO-1, SPO-2) in outflow facies outside of the caldera and at two sites within the caldera, one near the southern margin (SPL-1) and the other (SPL-2) less than 20 m above a quartz syenite body (Figures 1–3). Four sites in the middle member were spaced broadly throughout much of the vertical extent of the approximately 300-m-thick unit (Figure 3). The upper member was sampled at three sites above, and at three below, the prominent break in the upper member, to within about 200 m of the top of uppermost exposed tuff (Figure 3). At most sampled outcrops, strikes and dips of the tuff were measured using eutaxitic structures to within about 5°–10°. At two outcrops, where the foliation was poorly defined, nearby structural relations were used to determine the attitude of bedding to within about 10° in strike and $<10°$ in dip. At the sampled localities, no evidence was found for high-temperature remobilization or deformation of the tuff or for

TABLE 1. Summary of Paleomagnetic Results From Units of the Lake City Caldera: af Demagnetization

Site	Latitude °N	Longitude °W	N	Treatment, mT	D	I	D_u	I_u	Tectonic Correction: Strike, deg	Tectonic Correction: Dip, deg	α_{95}	k	Latitude	Longitude	δp	δm
Sunshine Peak Tuff																
Upper limit																
SPU-1	37.96	107.37	8	60	126	−58	144	−43	88	20S	5.5	101	48	181	6.0	8.1
SPU-2	37.92	107.35	9	60	97	−51	64	−68	220	25N	4.1	156	24	185	3.7	5.5
SPU-3	37.92	107.38	9	70	195	−74	187	−54	90	20S	7.9	43	55	323	9.0	11.9
SPU-4	37.92	107.42	7	40	207	−77	172	−61	58	20S	2.3	705	67	266	3.7	4.2
SPU-5	37.93	107.40	7	40	195	−73	171	−56	58	20S	3.5	297	58	269	6.2	6.6
SPU-6	37.92	107.39	5	60	198	−71	160	−54	40	25S	5.3	208	69	282	8.0	9.2
Middle unit																
SPM-1	37.96	107.37	8	20	97	−44	117	−34	85	25S	6.6	70	17	182	5.2	8.3
SPM-2	37.93	107.39	6	25	126	−67	142	−44	70	25S	2.0	1168	50	198	2.7	3.3
SPM-3	37.93	107.35	8	20	112	−63	···	···	0	0	3.3	277	40	193	4.1	5.2
SPM-4	37.93	107.40	7	40	136	−70	160	−49	90	25S	4.4	190	56	205	6.5	7.6
Lower unit																
SPO-1	38.05	107.24	7	40	93	−65	···	···	0	0	2.2	755	29	202	2.9	3.6
SPO-2	37.99	107.17	8	20	100	−63	···	···	0	0	2.7	424	32	197	3.3	4.2
SPL-1	37.92	107.46	9	30	101	−55	···	···	0	0	4.3	143	29	187	4.3	6.1
SPL-2	37.95	107.43	6	*	N and R directions				0	0						
Intracaldera lava flows																
QG-1	38.00	107.34	7	60	171	−52	···	···	0	0	4.7	169	81	129	4.4	6.4
QG-2	37.96	107.34	4	60	142	−40	134	−4	20	40S	9.8	88	54	149	7.1	11.8
QG-3	37.98	107.31	7	20	117	−48	···	···	0	0	3.2	362	38	173	2.7	4.2
QG-4	38.00	107.33	7	80	175	−36	···	···	0	0	4.1	223	71	88	2.8	4.8
QG-5	38.00	107.33	5	40	162	−44	···	···	0	0	4.8	254	70	129	3.8	6.0
QR-1	37.98	107.32	10	50	163	−72	···	···	0	0	3.4	200	68	228	5.3	6.0
Intrusions																
SY-1	37.98	107.41	9	*	21	65	···	···	0	0	11.4	21	72	54	14.9	18.4
SY-2	37.97	107.43	9	*	342	68	···	···	0	0	5.0	105	72	147	7.0	8.4
SY-3	37.95	107.43	8	*	N and R directions				0	0						
QM-1	37.99	107.36	6	*	32	75	···	···	0	0	16.0	19	58	40	26.9	29.3
RH-1	37.97	107.46	6	*	21	52	···	···	0	0	8.7	61	72	7	8.1	11.9

N, number of samples; Treatment: demagnetization level resulting in least dispersion or directions determined by vector subtraction; *D*, declination in degrees from true north; *I*, inclination in degrees, negative upward; *D*, *I* represent tilt-corrected directions if followed by values for D_u, I_u or in situ directions if followed by leaders, indicating tilt correction unnecessary. D_u, I_u in situ declination and inclination, respectively, for sites requiring tilt correction; α_{95}, semiangle of the cone of 95% confidence in degrees; *k*, precision parameter [*Fisher*, 1953]; Latitude, longitude are north latitude and west longitude of virtual geomagnetic pole (VGP), in degrees; δp and δm, semiminor and semimajor axes (in degrees), respectively, of the oval of 95% confidence surrounding the VGP.

*Directions determined by vector subtraction.

steep initial dips. Where sampled, the tuff has dips of 25° or less (Table 1).

Directions of natural remanent magnetization (NRM) are closely grouped for most sites; NRM intensities average 0.05, 0.90, and 0.17 A m^{-1} for the lower, middle, and upper members, respectively. Specimens of Sunshine Peak Tuff almost uniformly show univectorial decay in response to af demagnetization. Treatment at low peak fields (≤ 10 mT) removes from a few specimens a small normal component (northerly declinations and positive inclinations) that probably results from viscous overprinting, but higher fields (as much as 90 mT) isolate magnetizations having moderate to steep negative inclinations (Figure 4*a*). Mean directions of the lower and middle Sunshine Peak Tuff and of two sites in the upper member have easterly declinations (Table 1 and Figure 5). Of the latter, one site (SPU-1) is near the base of the upper member, and the other (SPU-2) lies directly above an apparently discontinuous bed of air fall ash within the top subunit of the upper member. In contrast, southerly declinations and steep negative inclinations were found for the remaining sites within the upper member (Table 1 and Figure 5).

Thermal demagnetization at low temperatures (≤ 200°C) removes normal components from some specimens, but continued heating to 680°C isolates remanent directions identical to characteristic directions defined by af demagnetization of specimens from the same site (Figure 4*b*). The results of thermal demagnetization indicate that magnetite is responsible for most of the remanence in the Sunshine Peak Tuff and that hematite, where present, carries the same remanent direction.

The presence of magnetite and hematite was confirmed by petrographic, thermomagnetic (saturation magnetization versus temperature), and IRM studies. Magnetite and hematite in the ash flow tuff occur in a wide variety of forms and associations that, to a great extent, appear to reflect high-temperature oxidation. Maghemite, a low-temperature alteration product of magnetite, is rare. Most samples contain magnetite as small relicts in large metatitanomagnetite grains that consist mostly of ilmenite lamellae and hematite, and as small (5–20 μm across) particles distributed throughout the groundmass. Moreover, tiny opaque grains (<1 μm diameter and too small for petrographic identification) that outline deformed and welded shards in many samples may also be magnetite [see *Reynolds*, 1977; *Geissman et al.*, 1983]. This inference is drawn from results of thermal demagnetization and from identification of magnetite in thermomagnetic and IRM analyses of a few samples which contain these tiny grains but

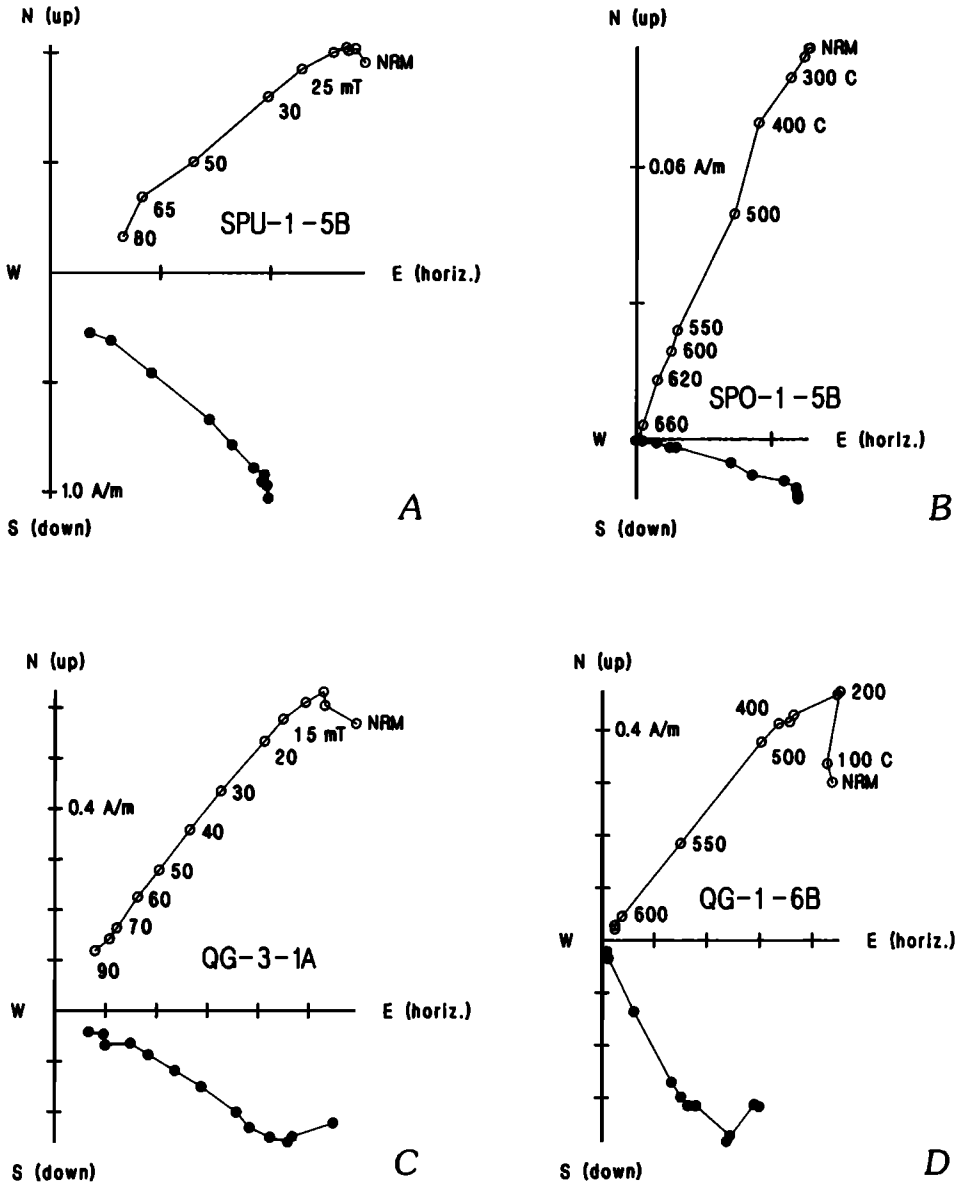


Fig. 4. Orthogonal vector diagrams depicting decay of magnetization in volcanic rocks of the Lake City caldera during progressive alternating field (in milliteslas) and thermal demagnetization (temperatures in degrees Celsius). (*a*) Sunshine Peak Tuff, upper unit; (*b*) Sunshine Peak Tuff, lower unit; (*c*) and (*d*) intracaldera lava flows. In this figure and in Figure 9, solid (open) symbols are projections of the magnetization vectors on the horizontal (vertical) plane and are plotted in in situ coordinates.

which lack larger magnetite grains. Hematite occurs predominantly as partial replacements of magnetite grains, as intergrowths within composite ilmenite-hematite grains, as segments in ilmenorutile-pseudobrookite, and as red translucent (extremely fine-grained) forms in the groundmass.

Thermomagnetic curves (Figure 6) reveal the presence of magnetite (Curie temperature of 580°C), titaniferous magnetite (Curie temperatures between about 520° and 560°C), and hematite in the Sunshine Peak Tuff. Curves are irreversible, as is demonstrated by cooling paths that return below the heating paths. For the most part, this may reflect partial oxidation of fine-grained magnetite during experimental heating, although in a sample from the upper unit (site SPU-1) the separation in paths was clearly augmented by the breakdown of maghemite to hematite.

IRM analyses are consistent with results of thermal demagnetization indicating that magnetite is the dominant contributor to the remanent magnetization in the Sunshine Peak Tuff. In the four specimens analyzed by IRM, the magnetizations were essentially saturated by applied fields of 0.2–0.3 T (Figure 7). Slight increases in magnetization at fields greater than 0.3 T reflect minor contributions by hematite.

Postcollapse Lava Flows and Quartz Latite Porphyry at Red Mountain

Postcollapse lava flows were sampled at five sites within the quartz latite of Grassy Mountain [*Lipman*, 1976*a*], and the quartz latite porphyry at Red Mountain that intrudes the flows was sampled from a dike at the northeast margin of Red Mountain (Figure 2). Field mapping indicates that the Grassy Mountain unit is composed of multiple flows, but the exact number is difficult to determine because the flow contacts are commonly obscure or incompletely exposed (R. Hon, unpublished data, 1983). Geologic and paleomagnetic data suggest

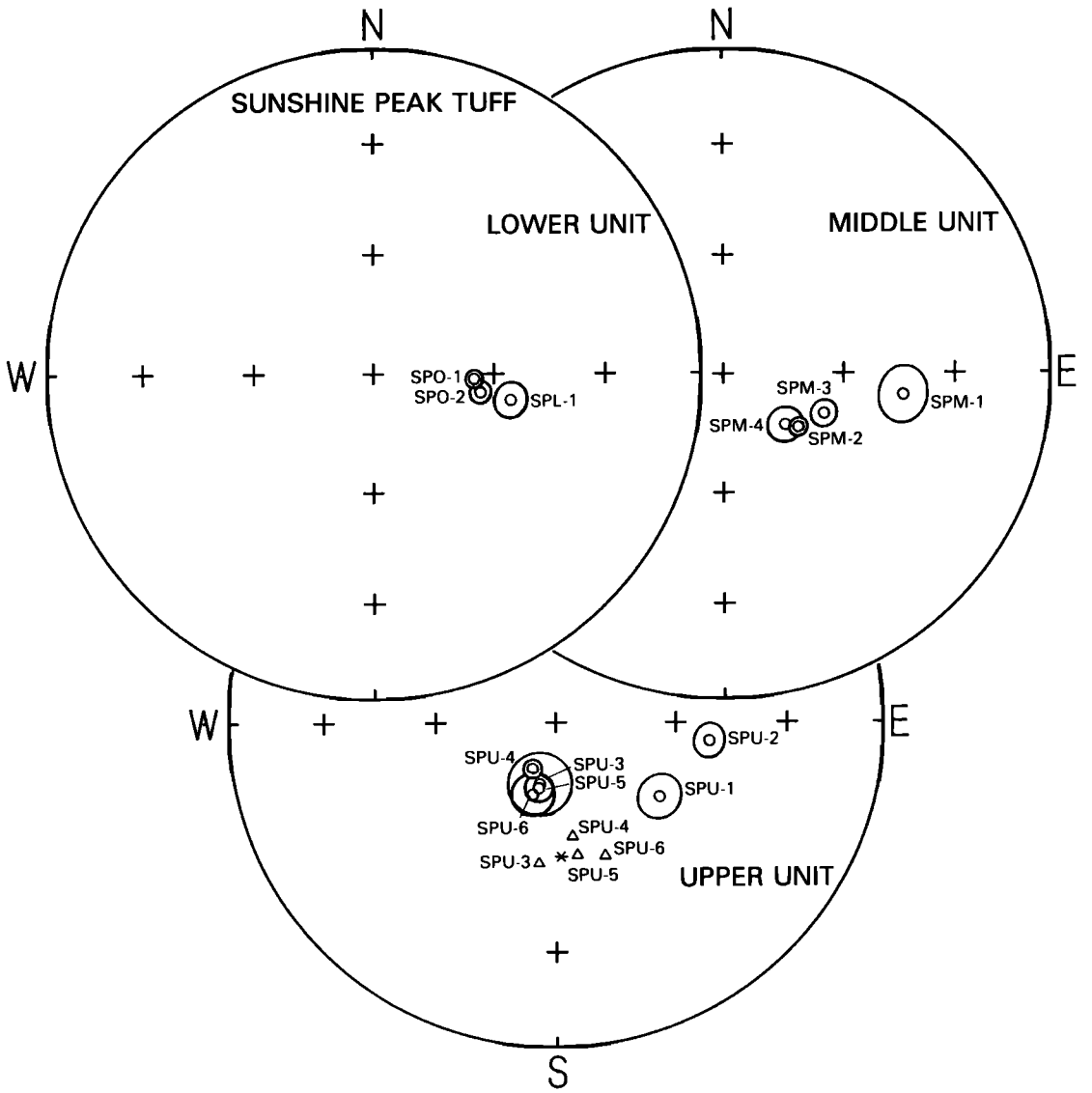


Fig. 5. Equal-area stereographic projections showing mean directions of magnetization for sites (excluding SPL-2; see Figure 11) in the three units of the Sunshine Peak Tuff (Table 1). In these and in following stereographic plots (Figures 8, 10, and 11), open symbols are projections from the upper hemisphere. Triangles are in situ mean directions that upon tilt correction give cluster of directions indicated for sites SPU-3, -4, -5, -6 (Table 1). Ellipses around site-mean directions are cones of 95% confidence [*Fisher*, 1953]. Asterisk denotes an average Miocene (~20 m.y.) direction for North America (from *Irving and Irving* [1982] and recalculated for Lake City geographic coordinates) and is shown in reversed polarity.

that the five sites in the Grassy Mountain unit represent at least four different flows; closely adjacent sites QG-4 and QG-5 may be in the same flow. A structural correction was applied to results from site QG-2 (Table 1), which is stratigraphically above steeply dipping (40°) sediments between two flows.

Magnetic properties of the quartz latites closely resemble those of the Sunshine Peak Tuff. NRM intensities average 0.63 A m^{-1}. In most specimens subjected to af treatment, magnetizations having southerly declination and negative inclination decay in a univectorial fashion. In others, af cleaning at low peak fields (<20 mT) removes a low-coercivity normal component before isolating at higher fields the characteristic reversed direction (Figure 4*c*).

Mean declinations of the Grassy Mountain flows are rather broadly distributed (from 117° to 175°), but mean inclinations (−36° to −52°) are not greatly different (Table 1 and Figure 8). The mean direction of the porphyry at Red Mountain (site QR-1) differs noticeably from those of the Grassy Mountain unit by virtue of its very steep (−72°) inclination (Table 1 and Figure 8).

Magnetic behavior of the quartz latites upon thermal demagnetization corresponds closely to results of af clearing. Normal components are eliminated by heating at 200°–400°C, and further heating progressively to 600°C isolates characteristic reversed magnetizations (Figure 4*d*). In samples having measurable remanence above the 600°C step, reversed components removed between 600°C and 680°C are nearly identical to those isolated over lower temperature intervals. These results point to magnetite as the main carrier and to hematite of deuteric origin as a minor carrier of remanence in the lava flows, as was found in the Sunshine Peak Tuff.

Magnetite in the lava flows and in the porphyry at Red Mountain occurs typically as relicts within altered titanomagnetite phenocrysts, and as smaller ($\leq$ 10 μm) equant grains partly replaced by specular hematite. Thermomagnetic and

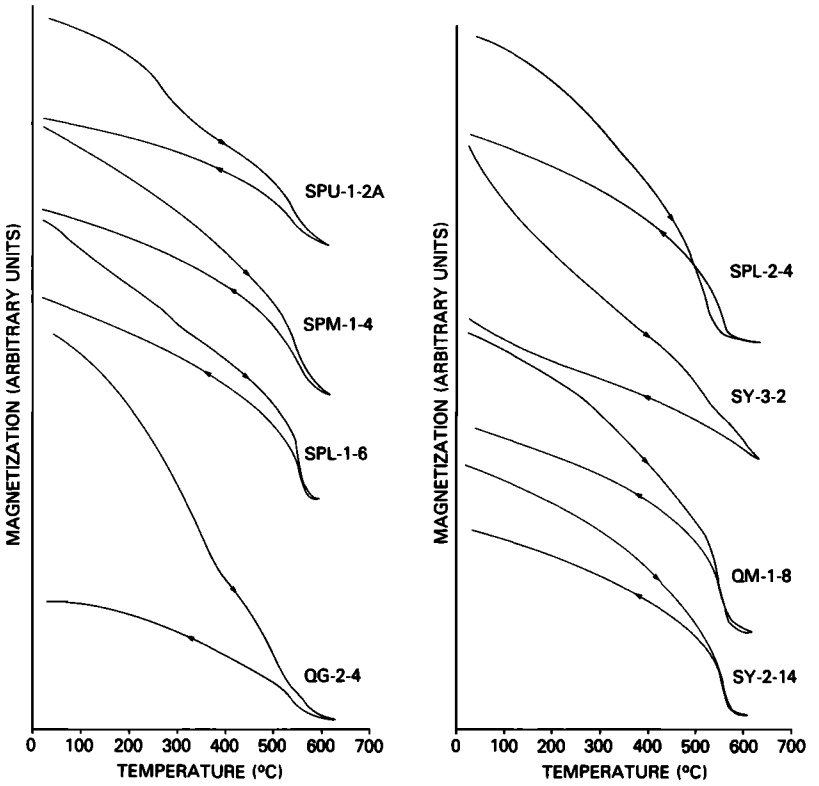


Fig. 6. Thermomagnetic curves (magnetization versus temperature) of rock chips (heated in air) from the upper, middle, and lower units (SPU, SPM, and SPL, respectively) of the Sunshine Peak Tuff, intracaldera lava flow (QG), and quartz syenite (SY) and quartz monzonite (QM) intrusions. Sample SPL-2-4 is from remagnetized tuff and sample SY-3-2 is from the underlying quartz syenite.

IRM-acquisition curves are dominated by the presence of magnetite (Figures 6 and 7).

Quartz Syenite, Quartz Monzonite, and Rhyolite Intrusions

The three sites in quartz syenite represent a range of depths within the upper parts of the resurgent intrusion. Site SY-3 was at the uppermost level in the body (within 5 m of the Sunshine Peak Tuff), site SY-2 was within 20 m of the superjacent tuff, and site SY-1 was about 200 m from the nearest known exposure of tuff (Figure 2). Although the depth represented by site SY-1 cannot be accurately determined, it is greater than those of sites SY-2 and SY-3. Sites QM-1 and RH-1 similarly are about 200 m or more from the nearest outcrops of megabreccia of the Sunshine Peak Tuff (Figure 2). Tectonic corrections for the intrusive rocks could not be determined and are most likely not required because field evidence suggests little, if any, structural tilting within the caldera after resurgence.

NRM directions for each site are scattered and NRM intensities of specimens even from the same site range broadly (e.g., 0.88×10^{-3} to 0.39 A m^{-1}). The af cleaning reduces NRM dispersion and isolates normal mean directions for sites SY-1, SY-2, QM-1, and RH-1 (Table 1 and Figures 9*a* and 10). The af treatment of specimens from site SY-3, however, revealed both negative and positive inclinations and thus implied the acquisition of remanence during times of both reversed and normal polarity fields.

Thermal demagnetization confirmed the presence of reversed components in specimens from site SY-3 and from the other sites. The reversed components are well defined over a broad range of unblocking temperatures in all specimens from the margin of the quartz syenite at site SY-3 and in most specimens near the intrusive margin at site SY-2. In contrast, reversed components are found rarely in samples (or are isolated over a limited range of temperatures) from deeper within the interior of the quartz syenite intrusions, and from the rhyolite (RH-1) and quartz monzonite (QM-1) intrusions.

In each specimen from site SY-3, heating to low temperatures (up to 300°-500°C) removed a well-defined normal component, whereas heating to higher temperatures (to 680°C) removed a reversed component (Figures 9*b* and 11). Mean directions of the normal and reversed components are $D = 7°$, $I = 71°$, $\alpha_{95} = 6°$, and $D = 186°$, $I = -45°$, $\alpha_{95} = 18°$, respectively. Magnetite and hematite also carry reversed magnetizations (or those having negative inclinations) in seven of eight thermally demagnetized specimens from nearby site SY-2 (Figure 10). These components outline a broad band or streak from a reversed direction (near the mean direction of site SY-3) to directions having westerly declinations. Such a streak may result from different amounts of superimposed normal and reversed magnetizations that are unresolved by demagnetization.

Thermal demagnetization of eight specimens from site SY-1, which is farther than sites SY-2 and SY-3 from the margin of the quartz syenite pluton, disclosed reversed magnetizations in only two specimens. In one of these, the reversed component ($D = 194°$, $I = -24°$) is carried by hematite, and in the other ($D = 235°$, $I = -17°$) it is carried by magnetite. Of the remaining specimens, four exhibited normal magnetizations during progressive demagnetization (Figure 10), whereas the others showed erratic, uninterpretable behavior. The reversed

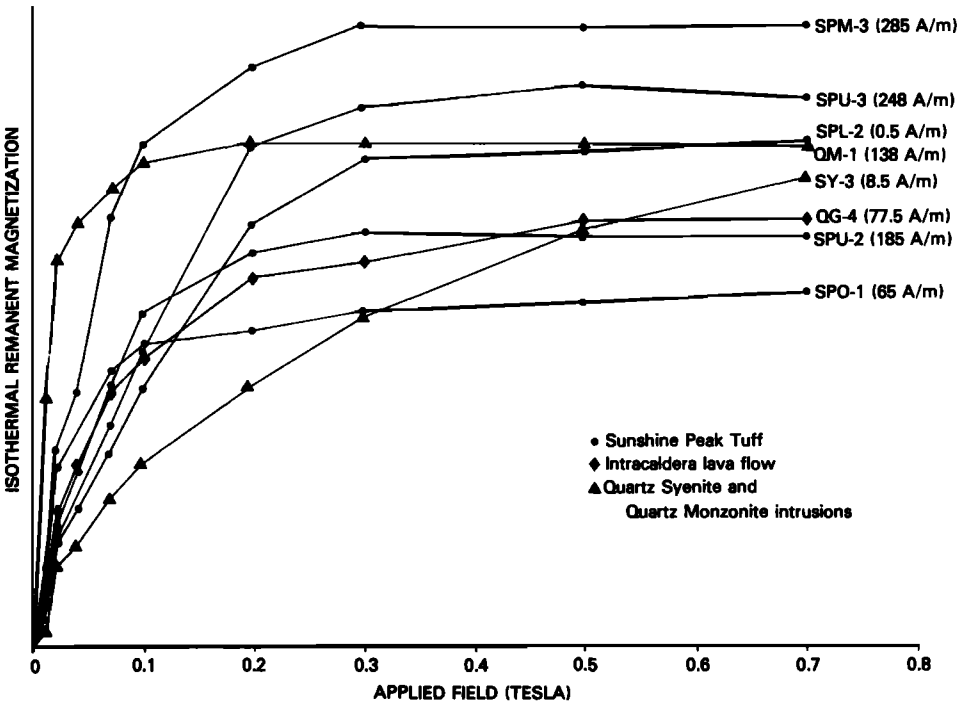


Fig. 7. Curves of isothermal remanent magnetization (IRM) versus applied external magnetic field for the Sunshine Peak Tuff (SPU, SPM, SPL, SPO; circles), intracaldera lava flow (QG; squares), and quartz syenite (SY) and quartz monzonite (QM) intrusions (triangles). Intensities of IRM after treatment at 0.7 are shown in parentheses.

components found in the quartz monzonite (site QM-1) are removed at temperatures greater than 600°C and thus are held in hematite. The mean direction of this reversed component is $D = 146°$, $I = -32°$, $\alpha_{95} = 14°$ ($N = 6$) (Figure 10). Components removed at $T < 600$°C are characterized by steep (~80°) positive inclinations. Only one of six specimens from the rhyolite plug had a reversed direction ($D = 176°$, $I = -53°$), removed from 400° to 600°C. Curiously, magnetite and hematite in each of the remaining five specimens (five samples) had unusual but well-defined directions having N-NE declinations (mean $D = 27°$) and negative inclinations (mean $I = -71°$; $\alpha_{95} = 6°$).

In a baked-contact test the lower unit of the Sunshine Peak Tuff from site SPL-2 exhibits magnetic behavior essentially identical to that of the underlying quartz syenite (site SY-3). Components in the tuff removed at low temperatures (below 400° or 500°C) and at high temperatures (400° or 500° to about 640°C) correspond closely to components removed from specimens of the intrusive body over similar temperature intervals (Figures 9c and 11). The mean direction ($D = 215°$, $I = -45°$, $\alpha_{95} = 14°$, $N = 6$ specimens from six samples) of the reversed components in the remagnetized tuff differs greatly from those of the lower unit at other localities.

Results of thermal demagnetization and petrographic, IRM, and thermomagnetic analyses indicate that remanence in the intrusions is carried dominantly by magnetite and that hematite contributes little to total remanent intensity, except in some samples from site SY-3. For the most part, magnetite occurs as grains (5–200 μm across) which exhibit different degrees of division by ilmenite-rutile and of replacement by hematite. The paucity of magnetite and an abundance of hematite in some polished sections of samples from site SY-3 are reflected by shallow slopes of curves of IRM acquisition for the same samples (Figure 7).

DISCUSSION

Sunshine Peak Tuff

A number of lines of reasoning indicate that each of the two main clusters of reversed site-mean directions of the Sunshine Peak Tuff, one cluster (nine sites) characterized by E-SE decli-

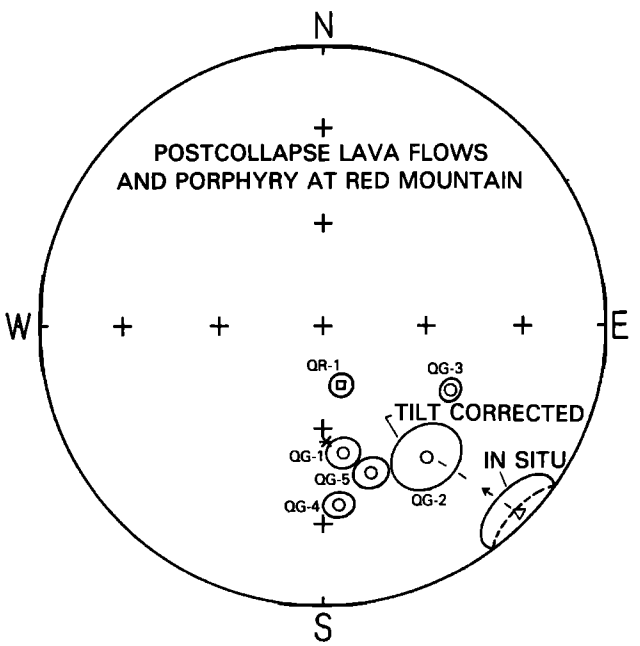


Fig. 8. Equal-area stereographic projection showing site-mean directions of postcollapse lava flows (small circles) and of the prophyry at Red Mountain (square). The inverted triangle represents the mean direction of the flow from site QG-2 before tilt correction (in situ). Asterisk denotes average Miocene direction as in Figure 5.

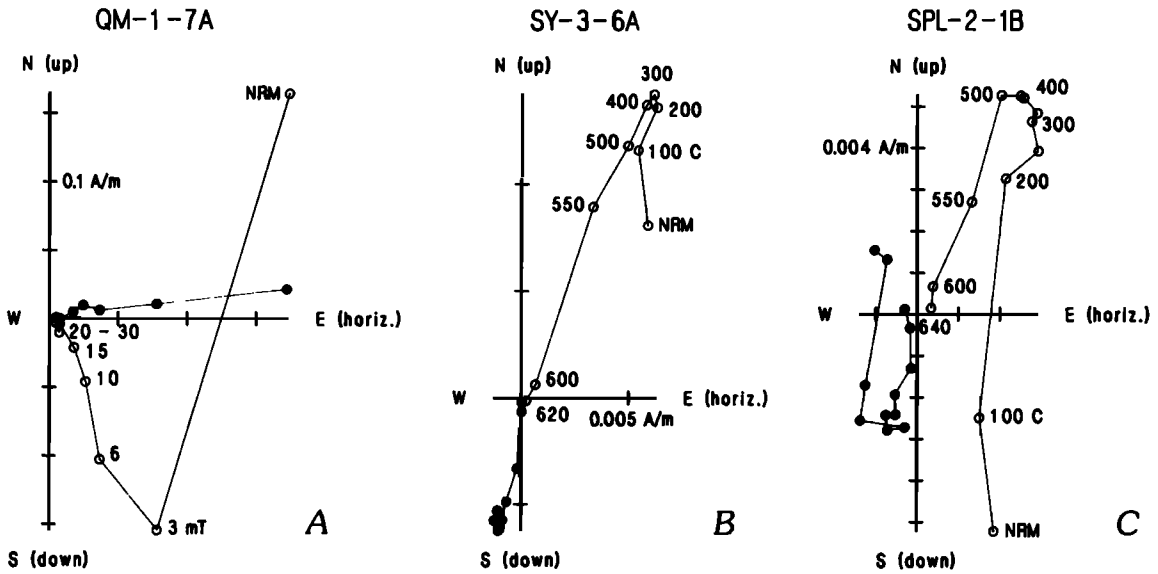


Fig. 9. Orthogonal vector diagrams depicting decay of magnetization during progressive af demagnetization of a specimen of quartz monzonite (Figure 9*a*) and during progressive thermal demagnetization of quartz syenite at site SY-3 (Figure 9*b*) and of the lowest unit of the Sunshine Peak Tuff at site SPL-2 (Figure 9*c*) that has been thermally remagnetized by the intrusion at site SY-3.

nations and the other (four sites) by southerly declinations and steep negative inclinations (Figure 5), represents a primary magnetization. The directional difference between these groups of magnetizations apparently reflects changes in paleomagnetic field directions during cooling and magnetization of the ash flow tuff. Primary magnetizations of the E-SE group are demonstrated by the similar mean directions of intracaldera tuff and of outflow tuff, the latter of which was clearly not affected by later hydrothermal and intrusive activity within the caldera. Moreover, these unusual reversed directions are unlike the reversed directions isolated in the intracaldera intrusions (Figure 10). The discordance between the E-SE reversed directions (which define a mean direction of $D = 107.7°$, $I = -60.3°$, $\alpha_{95} = 7.1°$) and the average Miocene direction for North America ($D = 178.1°$, $I = -56.1°$, $\alpha_{95} = 3°$, recalculated for Lake City geographic coordinates [*Irving and Irving*, 1982] (Figure 5)) is greater than that predicted by magnetic secular variation [see *Merrill and McElhinny*, 1983].

Mean directions composing the E-SE group, however, do not define systematic changes with stratigraphic order of sampled tuff. Some dispersion probably arises from uncertainties in tectonic corrections. Because these uncertainties are too small to account for most differences between the site-mean directions, an additional cause for the scatter (variations in the paleomagnetic field as the tuff cooled at different rates) seems likely.

A curious aspect of the paleomagnetic results is the large difference in declinations of the mean directions (Figure 5) found for sites within the upper unit. The two mean directions having easterly declinations were found in tuff that was chilled rapidly: one site (SPU-1) lies within 2 m of a thin sheet of collapse breccia at the contact between the middle and upper members, and the other (SPU-2) is directly above a discontinuous nonwelded ash bed within the top portion of the upper member (Figure 3). Of the four sites (SPU-3, -4, -5, -6) yielding southerly mean declinations, two were sampled above and two below the break that occurs about 300 m below the top of exposed tuff.

The grouping of the in situ mean directions of these latter four sites about an average Miocene direction for North America (from *Irving and Irving* [1982], recalculated for Lake City geographic coordinates; see Figure 5 and Table 1) and near the reversed direction for quartz syenite at site SY-3 (see Figure 10) raises the possibility that remanence in this part of

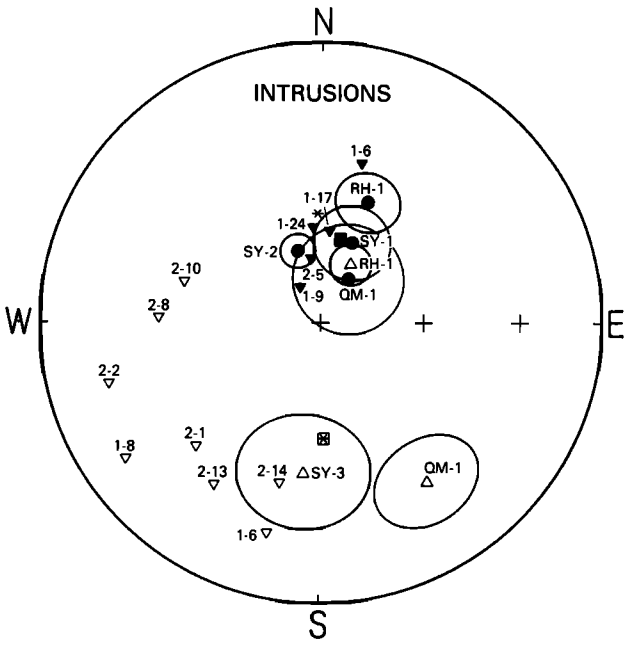


Fig. 10. Equal-area stereographic projection showing directions of magnetization for intracaldera quartz syenite (SY), quartz monzonite (QM), and rhyolite (RH) intrusions. Solid (open) symbols are projections from the lower (upper) hemisphere. Small circles indicate mean directions (surrounded by cones of 95% confidence) after progressive af demagnetization (Table 1). Triangles indicate directions of components removed during progressive thermal demagnetization. Mean directions for sites SY-3, QM-2, and RH-1 are surrounded by 95% confidence cones. Inverted triangles are directions of components removed from individual specimens (identified by number adjacent to symbol) from sites SY-1 (e.g., 1–6) and SY-2 (e.g., 2–13). Asterisk denotes average Miocene field (see Figure 5) shown in normal polarity. Asterisk inside the open square denotes average Miocene field in reversed polarity. Solid square represents direction of the present earth's field at Lake City.

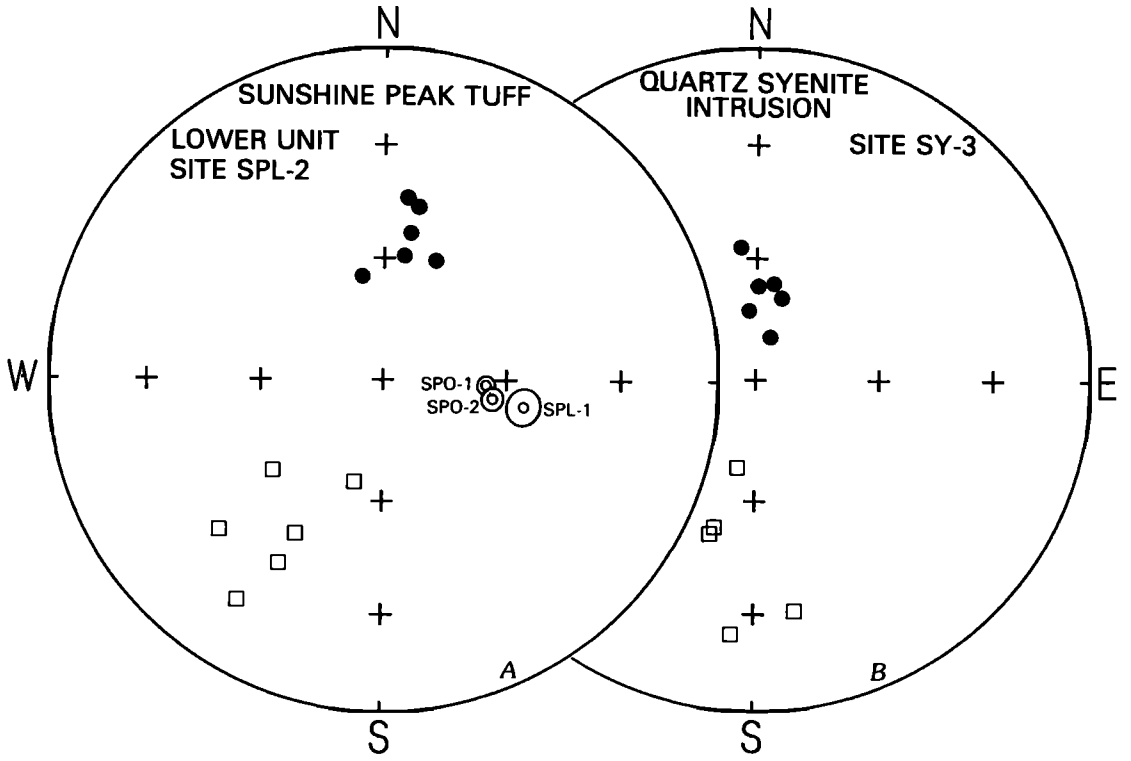


Fig. 11. Equal-area stereographic projection of components of magnetization removed during thermal demagnetization of specimens (*a*) from site SPL-2 in the lower unit of the Sunshine Peak Tuff and (*b*) from site SY-3 in the immediately underlying quartz syenite resurgent intrusion. Small solid circles represent components removed at low temperatures (typically $T \leq 400°C$); open symbols, components removed at higher temperatures (typically, $T \geq 400°C$). Also shown on Figure 11*a* are mean directions (small open circles) for the three other sites (SPO-1, SPO-2, SPL-1) in the lower unit of the Sunshine Peak Tuff (Table 1). Solid (open) symbols represent projections from the lower (upper) hemisphere.

the upper unit was reset by intrusive or accompanying hydothermal activity after or during the late stages of resurgent doming. For the following reasons, however, we believe that the remanence in the uppermost sites is primary and that tilt corrections are required (Table 1). First, nearby tuff lower in the section (e.g., sites SPM-2 and SPM-4, Figure 2) does not show evidence of remagnetization. Second, we found no evidence for greater or different alteration of tuff at these four sites than at the other two sites in the upper unit. Finally, between-site grouping for these four sites ($k = 130$, for in situ directions) is greatly increased after correcting for tilt ($k = 780$).

The paleomagnetic results indicate that the earth's field was reversed, but with an unusual easterly declination, during cooling of most of the Sunshine Peak Tuff. Parts of the upper unit, however, recorded a field direction that had shifted to a more southerly declination and a steeper inclination.

Postcollapse Lava Flows and Quartz Latite Porphyry at Red Mountain

Field relations demonstrate that the emplacement of the quartz latitic lavas spanned the time of resurgence; this conclusion is corroborated by the paleomagnetic results. As discussed above, the flow at site QG-2 predated resurgence. Upon tilt correction its site-mean direction moves to one closer to other site means (Figure 8). In contrast, the lava flow sampled at site QG-1 postdated resurgence because it lies unconformably on gently tilted and partly eroded Sunshine Peak Tuff. The flow at sites QG-4 and -5 has a remanent direction similar to that at site QG-1, and it also may have been emplaced after resurgent doming.

Three factors other than tilting by resurgence may have contributed to the observed scatter in site-mean directions of the quartz latites: (1) postresurgence structural rotations, (2) flowage of chilled portions of the flows during emplacement, and (3) different paleomagnetic field directions recorded by individual flows. Three sites (QG-1, QG-3, and QR-1) are bounded by or are close to mapped Quaternary block-slump faults [*Lipman*, 1976*a*]. Slight rotations along these faults of large blocks containing these sites (or rotations of blocks containing these or other sites that are unrelated to the slump faults) may partly account for differences in their mean directions. Such unrecognized rotations may explain the discrepancy in directions for nearby sites QG-4 and QG-5, which are probably in the same flow. Another possible source of dispersion is flowage of completely or partly chilled parts (particularly margins) of flows. This possibility is considered unlikely because the flows generally were sampled over sufficiently broad areas to incorporate different attitudes of flow banding and because it is unlikely that sites were near original flow margins, the flows having been deeply eroded. Moreover, within-site dispersion is small, and thermal demagnetization reveals single-component decay of magnetization. Large within-site disperison and (or) slightly different components [see *Hoblitt et al.*, 1985] would have suggested flowage-related rotations of chilled parts of flows. Of the three possible factors, directional variations in the earth's field during postcollapse volcanism is the most plausible to account for most between-site dispersion.

Quartz Syenite, Quartz Monzonite, and Rhyolite Intrusions

The paleomagnetic data strongly suggest that the quartz syenites possess reversed and normal partial thermoremanent magnetizations (PTRM) acquired as cooling of the intrusion spanned a field transition from reversed to normal. The observation that the reversed directions are carried by both magnetite and hematite in specimens from the same site (SY-3; Figure 10) implies that both minerals were magnetized at elevated temperatures under similar magnetic fields. It is thus unlikely that a secondary, low-temperature chemical remanent magnetization (CRM) can account for these reversed components. The resolution of reversed components and the uniformity of magnetization appear to be related to the distance from the intrusive margin and, consequently, to cooling rates of the intrusive rocks. The best resolution and most uniform behavior (all specimens changing polarity over nearly identical intervals of demagnetizing temperatures) are found at site SY-3, adjacent to the contact of quenched border-phase quartz syenite with the Sunshine Peak Tuff near the center of the caldera. Such behavior may reflect shorter times for cooling and thus for blocking of magnetizations relative to those for other sites.

All but one specimen from site SY-2 have negative inclination magnetizations, but these magnetizations define a streak from near the mean direction of site SY-3 toward the northwest. The large sampling area at site SY-2 (broader than that at SY-3), and a correspondingly wide range of cooling times, may explain this scatter of remanent directions. Quartz syenite from the greatest depths sampled in the intrusion (site SY-1) presumably cooled more slowly and thus acquired much of its magnetization later than did quartz syenite from the other sites. These conditions may account for the presence of only two reversely magnetized samples at this site.

Strong evidence for primary reversed components in the quartz syenite comes from a comparison of the paleomagnetic results of the hornfelsed Sunshine Peak Tuff at site SPL-2 with those of the immediately adjacent intrusion at site SY-3: Reversed components are isolated from both the tuff and the intrusion over nearly identical ranges of laboratory unblocking temperatures (Figures 9 and 11). The baked-contact test also provides evidence that at least some of the normal magnetization in the quartz syenite is a PTRM. The removal of normal components in both tuff (site SPL-2) and quartz syenite (site SY-3) over nearly the same unblocking temperatures strongly implies that the normal component in the tuff was imparted by thermal remagnetization during cooling in a normal field. This implication is strengthened by the observation that these normal components are removed at higher temperatures than are normal directions (probably overprints of viscous remanent magnetization (VRM)) in some specimens of tuff at other sites, including in tuff outside the caldera and far from intrusive contacts. Moreover, tuff at site SPL-2 clearly has not been pervasively overprinted by VRM; af cleaning shows no evidence of normal magnetizations in the specimens having reversed NRM directions.

It is not known whether the reversed directions found in quartz monzonite and rhyolite were acquired during the same polarity chron as those in the uppermost part of the quartz syenite. Because the reversed directions in the quartz monzonite (site QM-1) are carried solely by hematite, it cannot even be demonstrated whether these are primary (acquired at elevated temperatures) or chemical magnetizations. Only one sample from the rhyolite dike (site RH-1) is reversely magnetized. The origin of the anomalous directions (NE declinations, steep negative inclinations) found in the remaining samples is unknown. No reasonable slumping-related or tectonic rotation could account for these remanent directions if applied to typical normal or reversed directions. These remanent directions may approximate unusual paleofield directions, but a baked-contact test could not be made to assess this possibility. Although field relations indicate that the rhyolite probably predated the quartz syenite along the ring fracture zone, the time of rhyolite emplacement relative to cooling of the upper parts of quartz syenite in the center of the caldera is not known.

DURATION OF ERUPTION OF THE SUNSHINE PEAK TUFF

The similarity among the easterly mean directions found in each of the three members suggests that the entire Sunshine Peak Tuff was emplaced rapidly with respect to secular variation of the geomagnetic field. The tuff in thin exposures of the outflow facies was among the earliest erupted material, and it became chilled, and thus magnetized, earlier than did most or all of the intracaldera tuff [see *Riehle*, 1973]. For this reason, site-mean directions of the outflow tuff provide the best estimate of the paleomagnetic field direction shortly after (probably within a few years) eruption of the Sunshine Peak Tuff. Mean directions of two sites (SPM-3 in the middle unit and SPU-2 in the top subunit of the upper unit) are closely similar to the directions from the outflow tuff. It is unlikely that during cooling of the intracaldera tuff, the earth's field returned repeatedly to this anomalous direction; more likely, tuff at sites SPM-3 and SPU-2 became magnetized in a direction that had not changed much since the earliest ash flow eruptions. Uniformity of magnetic directions has also been found at the bottoms and tops of similarly thick sections of some other ash flow tuffs [*Geissman et al.*, 1982], even in one (the 2.0-m.y. Huckleberry Ridge Tuff) that cooled during a transition of the geomagnetic field [*Reynolds*, 1977].

Four sites (SPU-3, -4, -5, -6) in the upper unit, however, yield tightly clustered southerly directions that are 20°–40° from the other site-mean directions. These southerly directions are ascribed to a shift in the paleomagnetic field from easterly to southerly, as the tuff at these four sites cooled through magnetization blocking temperatures later than, for example, site SPU-2, as discussed above.

Additional information regarding the cooling of the Sunshine Peak Tuff is gained by comparing paleomagnetic dispersion of this unit with the dispersions of another ash flow tuff and of other cooling units in the San Juan volcanic field. The parameter used in these comparisons is the angular disperison of secular variation (as angular standard deviation S) measured among virtual geomagnetic pole (VGP) positions (calculated from site-mean directions) distributed about a mean VGP [*Doell*, 1970]. For the Sunshine Peak Tuff, S is 32.2°, with 95% confidence limits [*Cox*, 1969], $25.3° \leq S \leq 44.0°$. Such angular dispersion appears to be much greater than expected, even for a thick compound cooling unit. Although cooling of some interior parts of an ash flow sheet through a range of magnetization blocking temperatures (<580°C for magnetite) may take hundreds or a few thousands of years [*Riehle*, 1973], this is typically a short interval of time relative to paleomagnetic secular variation. Under these conditions one would thus expect low values of S because short cooling histories would record only a small proportion of the total field variation that may occur over periods greater than 10^4

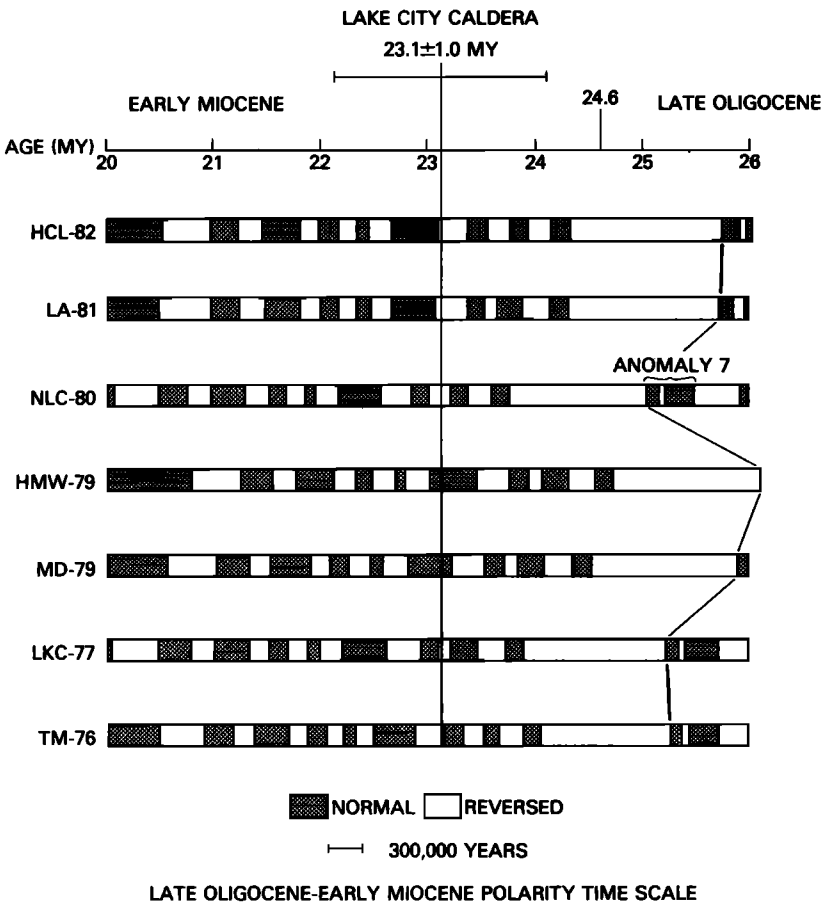


Fig. 12. Marine magnetic anomaly time scales for a portion of the late Oligocene and early Miocene (modified from *Ness et al.* [1980, Figure 2]). Time scale HLC-82 is constructed from Table 4.3 of *Harland et al.* [1982], and LA-81 is from Table 2 of *Lowrie and Alvarez* [1981]. The remainder are taken from *Ness et al.* [1980]. The bar illustrating 23.1 ± 1.0 m.y. at the top of the diagram represents the K-Ar age used for all igneous activity related to the Lake City caldera [see *Hon et al.*, 1983*a*].

years. This point is illustrated by the angular standard deviation of VGPs of the normally magnetized, 0.60-m.y. Lava Creek Tuff related to the Yellowstone caldera [*Christiansen*, 1979, 1982, 1984; *Reynolds*, 1977]. For the Lava Creek Tuff, S is 7.0° with 95% confidence limits, $5.5° \leq S \leq 9.6°$. (This result was calculated from 13 sites, including only two from outflow facies, in an attempt to balance this comparison.) When compared to volcanic rocks from the San Juan volcanic field representing long time periods (about 10^7 years), the angular dispersion of secular variation for the Sunshine Peak Tuff is found again to be greater than the disperison (23.9°) for 36 Oligocene cooling units [*Beck et al.*, 1977] and that (18.9°) for 12 Oligocene units and one early Miocene unit (the upper Sunshine Peak Tuff) given by *Tanaka and Kono* [1973]. Moreover, the angular dispersion represented by the Sunshine Peak Tuff greatly exceeds the predicted angular dispersions (16°–20°) for secular variation of the middle Tertiary field at Lake City [see *Cox*, 1970; *Merrill and McElhinny*, 1983].

Angular dispersion of the Sunshine Peak Tuff thus is anomalously large, even when considered in relation to long-term magnetic secular variation. Some of the angular dispersion probably results from uncertainties in tilt corrections, and some may arise from flow-induced grain rotation [see *Rosenbaum and Snyder*, 1985]. Nevertheless, cumulative contributions from these sources are not sufficiently large to account for all of the dispersion. An additional explanation for the high dispersion is that magnetization of the Sunshine Peak Tuff recorded a geomagnetic field that was changing direction, perhaps at an abnormally fast rate. However, the directional record of this field is too limited, and other evidence (such as paleointensity values) is lacking, to support firmly our earlier speculation [*Reynolds et al.*, 1983] that the Sunshine Peak Tuff cooled during a transition or excursion of the earth's field.

The closely similar directions from the bottom and near the top of the Sunshine Peak Tuff are consistent with petrologic, geochemical, and field evidence that point to rapid eruption of the entire ash flow tuff [*Hon et al.*, 1983*a*, *b*]. The paleomagnetic data, however, do not refine or improve estimates for rates of eruption of ancient ash flow cooling units [*Christiansen*, 1979]. These estimates are based partly on field observations (e.g., cooling characteristics and lack of erosion or of lava flows within thick ash flow sheets) and partly on analogies to voluminous late Pleistocene and historical pyroclastic flows, whose eruption times can be determined rather accurately from study of tephra layers [*Ninkovich et al.*, 1978; *Bacon*, 1983] or from eyewitness accounts [*Self et al.*, 1983; *Hildreth*, 1983]. These observations and analogues suggest

TABLE 2. Durations of Caldera Development

Caldera	Volume of Ash Flow Tuff, km³	Resurgence Time, years		Duration, Years	Age, Ma	References
		Minimum	Maximum			
Jemez Mountains caldera complex						
Toledo	>200	unknown	unknown	300,000	1.4	*Doell et al.* [1968], *Smith and Bailey* [1968], and *Smith* [1979]
Valles	>200	<100,000	150,000	<800,000	1.1	
Calabozos						
Early	250–300	unknown	unknown	150,000	0.3	*Hildreth et al.* [1984]
Late	175–250	<20,000	<50,000	150,000	0.15	
Long Valley	600	<80,000	<200,000	<700,000	0.70	*Bailey et al.* [1976]
Calderas of the Yellowstone Rhyolite Plateau						
First cycle	>2500	unknown	unknown	<700,000	2.0	*Christiansen* [1979, 1982, 1984] and *Smith and Christiansen* [1980]
Second cycle (Henry's Fork)	>280	no resurgence		<300,000(?)	1.3	
Yellowstone	>1000	<5,000–10,000(?)	<70,000	<600,000	0.60	
Lake City	300–400	unknown	unknown	<300,000	23.1	*Hon et al.* [1983*a*] and this paper

that many large calderas collapsed, and their associated ash flows erupted, in times of the order of months, weeks, or even days.

DURATION OF THE LAKE CITY CALDERA CYCLE

The lifespan of the caldera (from collapse to emplacement and cooling of the resurgent intrusion through magnetization blocking temperatures) is estimated by considering the paleomagnetic and radiometric data in conjunction with polarity time scales for the early Miocene. The paleomagnetic and radiometric data permit the explanation that the reversed directions found for the Sunshine Peak Tuff, for the intracaldera lava flows, and from the margins of the quartz syenite intrusions record a single interval of reversed polarity during early Miocene time. It is also possible that the rocks of the Lake City caldera acquired reversed magnetizations over a long time span ($\leq$2 m.y.) at times of different reversed polarity chrons.

If the reversed directions were acquired during one reversed polarity chron, the time interval of events associated with the development of the Lake City caldera is constrained by durations of reversed polarity intervals that occurred from ~22 to 24 Ma. All polarity time scales constructed for the early Miocene, the seven most recent of which are shown in Figure 12, illustrate numerous normal and reversed polarity chrons of brief duration between 20 and about 24 Ma [see *Harland et al.*, 1982; *Ness et al.*, 1980; *Lowrie and Alvarez*, 1981]. Within the interval 22–24 Ma, individual reversed polarity chrons are depicted as having lasted about 300,000 years or less. All polarity time scales show a long (~1.4 m.y.) reversed polarity chron following anomaly 7, a major reference point on the scales. In two times scales (NLC-80 and LKC-77) part of this long reversed interval overlaps the analytical uncertainty of the age of units within the Lake City caldera, but this overlap represents slightly less than 300,000 years in both scales. Assuming then that the characteristic magnetizations were acquired during a single reversed polarity chron, the time for development of the Lake City caldera, from eruption of the ash flow tuff to emplacement of the resurgent intrusion, was less than about 300,000 years. This estimated lifespan seems reasonable in view of the size and complexity of the Lake City caldera relative to sizes and complexities of much younger calderas, for which lifespans of the order of several 10^5 years (Table 2) [*Christiansen*, 1979] have been established by radiometric dating.

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K. Hon, M. R. Hudson, and R. L. Reynolds, U.S. Geological Survey, Box 25046, MS 964, Denver, CO 80225.

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